



1. EVENT OVERVIEW

OPERATION: REBUILD FROM CHAOS is an elite, high-intensity machine learning and reverse-engineering competition organised by NJACK ML as part of the NJACK Summer of Code (NSOC) programme. Operatives are dropped into a digital scenario where a trained deep residual neural network has been shattered into unlabelled weight matrices and bias tensors. Your mission is to apply mathematical forensics to reconstruct the network and recover the core classifications.

This operations exercise has been designed to evaluate:

- Deep Neural Network Topologies: Structural mapping of parameters and connections.
- Reverse-Engineering Diagnostics: Analytical heuristics of trained weight variables.
- Combinatorial Optimization: Search algorithms, matching strategies, and graph reassembly.

2. FORENSIC MISSION & OBJECTIVES

A trained residual network model classification head has been compromised. All N constituent layer tensors have been separated into a chaotic collection (piece_0.pth to piece_N.pth). Your objective is to:

- 1. Identify Component Types — Segregate the layers based on their weight tensor shapes.
- 2. Match and Pair Components — Correctly pair input projections with their corresponding output projections.
- 3. Sequence Residual Blocks — Discover the precise sequential order of the paired blocks from 1 to K.
- 4. Restore Grid Logits — Reassemble the original architecture to match predictions and recover classifications.

3. THE THREE PHASES OF ALIGNMENT

Phase	Description
Phase 1 — Distribution	Operatives receive the workspace package containing the scrambled layers, calibration data, and baseline code.
Phase 2 — Reconstruction	Operatives run exploratory analysis, develop pairing strategies, map the block sequences, and locally optimise their models.
Phase 3 — Submission	Teams package their automated pipelines, reconstructed weights, submission coordinates, and technical report.

4. COMPILER REPOSITORY STRUCTURE & CLONE GUIDE

All assets are tracked under a local workspace repository. Operatives should initialise their local grids as follows:

```
# Clone the central NSOC challenge repository
git clone https://github.com/chocopie247/OPERATION-REBUILD_FROM_CHAOS
cd OPERATION-REBUILD_FROM_CHAOS

# Spawn your dedicated local Team branch (Do NOT push to main!)
git checkout -b team-[your_team_name]
```

Path	Description
data/pieces/	Scrambled weight fragments (piece_0.pth to piece_N.pth)
data/historical_data.csv	Calibration alignment telemetry
samples/sample_submission.csv	Standard submission configuration layout
starter_kit.ipynb	Guided interactive forensic notebook

5. SECURED TELEMETRY (DATASET ANALYSIS)

The provided calibration dataset serves as intermediate alignment coordinates to verify your reassembly pipeline. The primary file is data/historical_data.csv. It represents representative input samples processed by the original model, capturing:

- Input feature dimensions: pixel_0 to pixel_X
- Predicted target logits: pred_logit_0 to pred_logit_Y
- Target class labels

6. DISMANTLED NEURAL CORE (MODEL COMPONENTS)

The N scrambled files must be classified strictly based on their weight tensor shapes:

Component	Qty	Shape	Operational Role
Front Projection (proj)	1	(D_latent, D_in)	Maps input space dimensions down to latent space features.
Classification Head (last)	1	(D_classes, D_latent)	Classifies latent features into output classes.
Input Projections (W_in)	K	(D_sub, D_latent)	Projects intermediate features up to a sub-block space.
Output Projections (W_out)	K	(D_latent, D_sub)	Projects sub-block features back down to latent space.

Residual Block Equation — each block k projects latent features x using the residual mapping:

Block_k(x) = x + W_out @ ReLU(W_in @ x + b_in) + b_out

7. ALLOWED FILE FORMATS

- Approved: .py .ipynb .csv .pth .txt .json
- Prohibited: Compiled executable binaries or external model weights.

8. TEAM RULES

- Team Size: 1 to 3 registered operatives per team.
- Independence: Complete isolation is required. Inter-team collusion will lead to immediate disqualification.

9. ARSENAL POLICY

- Allowed Tools: Standard Python libraries (PyTorch, SciPy, Scikit-Learn), reference manuals, and academic papers.
- Prohibited Tools: Web-based black-box solving APIs or sharing private logic blocks with other teams.

10. SUBMISSION

Teams must submit a compressed package named [team_name].zip containing:

- solution.py — Fully automated reassembly script
- final_model.pth — Reconstructed model weights
- submission.csv — Reassembled indices spreadsheet
- report.pdf — Exhaustive technical report

11. AFTER-ACTION REPORT

Your Technical Report must cover the following core outlines:

- Introduction
- Methodology
- Implementation Details
- Experiments / Results
- Challenges Faced
- Key Learnings

12. SCORE GRID & EVALUATION CRITERIA

Scores are calculated as:

Final Score = 0.70 × Model Score + 0.30 × Report Score		
Evaluation Area	Marks	Target Metric / Criteria
Model Logits MSE	25	MSE on intermediate classification logits
Reconstruction Quality	25	Correct layer pairing and sequential reassembly
Runtime & Efficiency	10	Speed and computational footprint of the solver
Technical Explanation	10	Mathematical depth and clarity in the report
Methodology & Innovation	8	Rigor and uniqueness of the pairing and search strategies
Experimental Analysis	6	Quality of testing, parameter logs, and performance curves
Presentation & Clarity	4	Document structure and readability
Reproducibility	2	Cleanliness of configuration guides and scripts
TOTAL	90	

Note: Model Logits MSE and Reconstruction Quality carry equal weight (25 marks each).

13. VERIFICATION & JUDGING DISCRETION

JUDGING PANEL RULE

The final evaluation, grading decisions, score validation, and official rankings rest solely at the absolute discretion of the NSOC organising team. The decisions of the judging panel are final and absolute.

Submissions utilising hardcoded indices, memorised mappings, or bypassing the core reassembly logic will fail verification and receive a score of 0.0.

14. STABILISATION TIE-BREAKERS

Submissions are evaluated against an isolated hidden dataset to prevent overfitting. If scores are equal, the following automated tie-breaking hierarchy applies:

Priority	Criterion
1st	Lowest Logits MSE
2nd	Highest Pairing Accuracy
3rd	Exact Sequence Match
4th	Least Forward Evaluations
5th	Earliest Submission Timestamp

15. RED LINES

Violations of the following will result in immediate disqualification:

- Plagiarism or copying another team's solution
- Coordinate sharing or leaking private logic blocks
- Dataset tampering or manipulation of calibration files
- Proxy registrations or misrepresentation of team members

16. NSOC PHILOSOPHY

The NSOC Command prizes mathematical elegance, creative problem solving, and computational efficiency. The spirit of this challenge is to push the boundaries of analytical reverse-engineering — not to find shortcuts around it. Operatives who demonstrate genuine ingenuity in their methodology will be recognised beyond the score alone.